

BEDSIDE HEMODYNAMIC MONITORING IN THE INTENSIVE CARE UNIT

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"No monitoring device can improve patient-centered outcomes unless it is coupled to a treatment that improves outcome."

Michael Pinsky

INTRODUCTION

The Danish anesthetist Bjørn Ibsen first pioneered the intensive monitoring of critically ill patients during the poliomyelitis epidemic in the early 1950s.² Yet it was not until the organization of coronary care units in the 1960s that hospitals formally adopted comprehensive monitoring of acutely ill patients safely outside the operating room setting. These units (much like today) offered continuous monitoring and aggressive interventions to improve survival in patients with acute myocardial infarction. The eventual application of this model to the care of noncardiac critically ill patients lent toward its incorporation into the core principles of intensive care medicine and set the standard for patient care in subsequent years.

The word "monitor" derives from the Latin word *monere*, meaning "to advise" or "to warn." In 1984 The Monitoring of The Critically Ill Conference³ expanded on this concept to include "making repeated or continuous observation or measurements of the patient, his or her physiological function, and the function of the life-support equipment, for the purpose of guiding management decisions, including one to make therapeutic interventions, and an assessment of those interventions."³ Though others were later introduced, this definition highlights the intimate relationship between patients (humans) and monitors (machines). It is from this foundation that hemodynamic monitoring now stands as a central foundation of clinical practice in the intensive care unit (ICU).

In the modern era intensive care medicine is now nearly synonymous with intensive cardiovascular monitoring. Still it remains worthwhile to mention that the elemental task of effective hemodynamic monitoring is the early detection of cardiovascular insufficiency and its clinical presentation as "circulatory shock." Shock is a life-threatening condition. It results in an uncoupling between tissue oxygen supply and demand, impairs cellular oxygen utilization, and leads to multiorgan failure if untreated. Common physical assessments for the presence and degree of clinical shock lack the immediacy and specificity needed to make accurate care decisions. By the time that common signs and symptoms of shock (such as hypotension, poor renal output and skin perfusion, and mental status changes) are present, important opportunities to intervene on a patient's behalf are missed. Therefore, early recognition of the syndrome of shock is paramount to acceptable outcomes that are achievable only with close mechanized monitoring of the hemodynamic system.

High fidelity and quality monitoring require defining specific parameters that accurately represent differences between healthy and diseased states. Individualized monitoring parameters must also cater to the specific needs of specific patient populations. For example, circulatory shock in post cardiac arrest patients, or patients with hepatic failure, or patients with recent neurological injury require different therapeutic approaches; the choice of monitoring modality is dependent on these needs. Disease progression also dictates prescribed monitoring techniques. Providers using less invasive methods for patients in mild cardiovascular insufficiency states must deftly escalate to more invasive monitoring for patients progressing through their disease course. Finally, it is important to highlight that process-specific monitoring methods effectively influence therapeutic strategies and allow for a basic understanding of cardiovascular physiology to guide bedside management decisions.

INVASIVE METHODS OF HEMODYNAMIC MONITORING

Arterial Blood Pressure Monitoring

Arterial blood pressure is an essential hemodynamic parameter in the ICU and reflects the interplay between vasomotor tone and cardiac output. Noninvasive blood pressure measurement by blood pressure cuff is, of course, a cornerstone of traditional patient vital sign assessments. Invasive arterial catheterization, a direct measurement of arterial blood flow, extracts data from the continuous monitoring of blood pressure. The arterial line pressure transducer setup links an independent catheter inserted into an accessible artery with a transducer via a fluid-filled tubing system. This allows for the conversion of pressure changes within the transducer into electrical signals that are displayed on a monitor with a waveform accompanied by numerical values. As with other monitoring modalities, this transducer system requires

careful instruction in meticulous priming, dampening assessment, zeroing, and leveling to ensure the interpretation of accurate data. With proper interpretation, the waveform tracing from the invasive arterial line can yield a treasure of clinical information regarding cardiovascular health. It offers nuanced and real-time insights into the cardiac cycle, including measurements of the systolic and diastolic blood pressure, mean arterial pressure (MAP), stroke volume, and pulse pressure.

The arterial pressure waveform represents a systolic and diastolic flow of blood through the cardiovascular system with each cardiac cycle (Fig. 4.1). It starts with a systolic upstroke after ventricular depolarization, manifested as the R wave on the electrocardiogram. Ventricular contraction results in a steep and positive deflection in the arterial blood pressure waveform culminating in peak systolic pressure. This segment is dependent on both the rate of change in left ventricular pressure and aortic valve integrity as different flow rates will create different slopes. For example, a patient with severe aortic stenosis has a delayed systolic upstroke segment. The decay in pressure after peak systolic pressure represents a decrease in stroke volume from an emptying left ventricle as it approaches diastole. The dicrotic notch, or “incisura,” occurs during this pressure decay and represents a brief, but abrupt, increase in blood pressure from aortic valve closure. This marks the end of systole. Arterial reservoir pressure and elastic recoil of the large arteries, in addition to peripheral vascular resistance, contribute to the gradual decline in blood pressure until it reaches the end of diastole. Peripherally vasodilated patients (as seen in individuals with hypovolemia, sepsis, cirrhosis, arterial venous malformations, or dialysis fistula placement) have a low systemic vascular resistance that showcases a sharp decline in the arterial waveform. Conversely those with a high systemic vascular resistance present with a rather shallow downstroke of the arterial waveform.⁴

Location of catheter placement greatly affects waveform characteristics. Canalization for the continuous monitoring of blood pressure usually takes place in the radial, brachial, or femoral artery. Peak systolic pressure is dependent on the changes in impedance across the vascular tree. Because compliance of the central arterial circulation decreases as blood moves more distantly through the divisions of the arterial tree, distal amplifications in blood pressure measurement occur further away from the aortic root. As a result, there is an increase in systolic blood pressure and a simultaneous decrease in diastolic blood pressure with little to no net effect on MAP in more distal vessels. A latent (or absent) dicrotic notch is seen the further a catheter is placed away from the aortic valve.

Perhaps the most coveted datum obtained from invasive measurement of arterial blood pressure is the MAP. Defined as the average blood pressure during a single cardiac cycle, MAP is a surrogate marker of organ perfusion pressure and greatly influences the autoregulation of blood flow. MAP incorporates principles of cardiac output, systemic vascular resistance, and venous return and is therefore a “holistic” marker of functional cardiovascular health. In the invasive measurement of blood pressure with an arterial catheter, MAP is the geometric mean area under the blood pressure waveform, while in noninvasive blood pressure cuffs it is mathematically obtained from the measured systolic and diastolic pressures. MAP targets are generally recommended by most professional guidelines to direct resuscitative efforts (particularly since they are often resistant to common bedside monitoring errors such as transducer dampening).⁴ That said, definitive MAP goals are elusive and vary across different guidelines. Studies comparing MAP targets from 65 to 85 mm Hg show no clear mortality differences in the general population.⁵ Paradoxically, targeting higher MAP goals is associated with increasing vasopressor use and the development of untoward cardiovascular effects.⁶

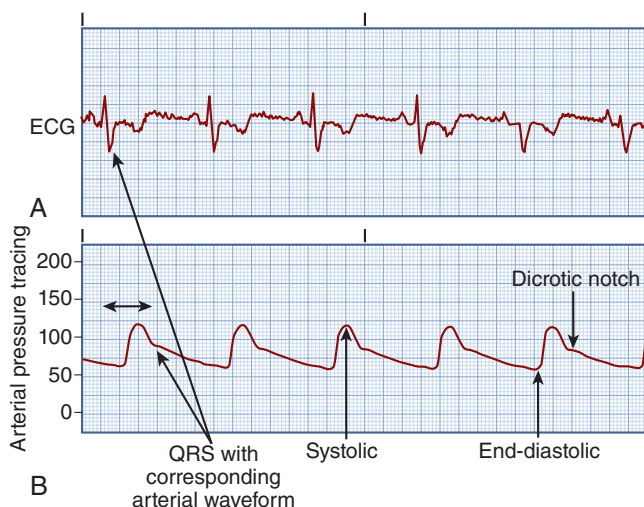


Fig. 4.1 Representative waveform obtained from an arterial line tracing. (Modified from Urden LD, Stacy KM, Lough ME. *Critical Care Nursing: Diagnosis and Management*. 6th ed. St Louis; 2010).

Pulse pressure is another important hemodynamic parameter of arterial pressure obtained from invasive arterial monitoring. It is the difference between systolic and diastolic pressure and represents a pulsatile component of blood pressure (as opposed to the static “point in time” variables represented by MAP). Pulse pressure is proportional to the stroke volume and inversely proportional to aortic compliance. The clinical application of this phenomenon is appreciated when the presence of a narrow pulse pressure indicates a decreased stroke volume (as is the case in cardiogenic shock, aortic stenosis, and various forms of obstructive shock) or a widened pulse pressure that represents an increased aortic compliance or stroke volume (as in hyperdynamic sepsis, anaphylaxis, and thyrotoxicosis).⁴

Monitoring of Cardiac Output With the Pulmonary Artery Catheter

Despite the abundance of data generated from invasive arterial monitoring, blood pressure is, at best, an acceptable surrogate marker for cardiac output. Cardiac output is the major determinant of oxygen delivery and a contributor to end-organ perfusion pressure. The direct monitoring of cardiac output, however, varies in its method of measurement, invasiveness, reliability, and user dependency.

There is no greater gold standard for the assessment of cardiac output than the invasive placement of the pulmonary artery catheter (PAC). Accurate placement of a PAC offers a package of data including intracardiac chamber pressures, the ability to directly assess cardiac output, and mixed venous oxygen saturation. Once ubiquitous, PAC use in critical care units decreased precipitously with publications indicating a lack of clear benefit in hospital outcomes.⁷ The introduction of noninvasive methods to measure cardiac output further reduced rates of PAC placement. As a result there is now a potential knowledge deficit in the interpretation of PAC waveforms, which may lead to improper hemodynamic profiling.

As described in Chapter 2, the PAC varies in length (60–110 cm) and caliber (4–7 French). Cardiac output is measured by the thermodilution method based on the Stewart-Hamilton equation or the Fick (direct or indirect) method based on the arteriovenous concentration difference of oxygen (see Chapter 3). Choice of PAC placement location in the catheterization laboratory is usually straightforward and dictated by clinician preference and patient clinical needs. The same holds true for the location of PAC placement in the ICU, but selection is also complicated by a lack of readily available fluoroscopic assistance and a long-term need for clinical monitoring. In the ICU setting, the PAC is commonly inserted into the internal jugular vein (potentially leading to inadvertent carotid arterial puncture), subclavian vein (technically more challenging and potentially causing pneumothorax), or femoral vein (associated with higher infection risk and difficulties with daily patient care).⁸ PAC use increases the risk of serious complications in the intensive care patient population such as thrombosis, arrhythmias, infections, and difficulties with insertion.⁹

PAC placement is indicated to distinguish cardiogenic from noncardiogenic shock when the clinical examination or noninvasive diagnostic methods are unclear. As the delivery of intravenous fluids is not without its own complications, information obtained from a PAC can also assist with establishing volume resuscitation goals. PAC placement is also indicated to assist with the diagnosis and guidance of therapeutic intervention in patients with advanced heart failure, pulmonary hypertension, complications of acute myocardial infarction, or mechanical circulatory support.

Accurate hemodynamic profile analysis is essential to diagnose different subsets of shock. The derived variables from PAC can assess in describing cardiac function, vasomotor tone, and pulmonary vasomotor tone, in addition to oxygen delivery and metabolism. Consensus guidelines recommend the implementation of less invasive modalities such as echocardiographic assessment before using a PAC and obtaining a solid understanding of patient hemodynamic physiology, pathophysiology, and profiling for adequate interpretation and intervention at the bedside once placement is pursued.¹⁰

Monitoring of Central Venous Pressure

Measurement of central venous pressure (CVP) may serve as a surrogate marker for cardiac output, and its use is considered when PAC placement is not indicated or unavailable. Transduced off the pressure tracings generated from a central venous line well-placed in the vena caval system, CVP is often described as a static hemodynamic parameter of right atrial pressure (Fig. 4.2). Appropriate CVP measurement is challenging in a vasodilated and volume underloaded (i.e., septic) patient as respiration can cause violent swings in intrathoracic pressure that are often more dramatic with the use of positive-pressure mechanical ventilation. By standardized consensus, CVP measurements are obtained at the end of expiration to avoid the transmural influence of pleural pressure on CVP values.¹¹

CVP use in critical care has fallen out of favor due to its poor numerical associations with preload, volume responsiveness, and cardiac output.¹¹ That said, CVP still finds value as a method of monitoring the cardiovascular system. Specifically, it can still serve as a marker of cardiac output—not for the absolute numerical value it generates—but as a relative value trended over time. When correlated with other points of clinical data (such as MAP, vasopressor requirements, and/or renal output), CVP provides valuable insight into patient hemodynamics. High CVP can occur due to decreased cardiac function or increased venous return as in pulmonary embolism and decompensated heart failure. Low CVP indicates decreased venous return due to volume depletion or increased venous resistance.

The waveform generated by CVP monitoring has the same characteristics as the right atrial waveform obtained from PAC placement. Generally speaking, understanding the fundamentals of CVP waveform analysis increases insight into patient's clinical condition. The absence of an a-wave indicates loss-of-atrial contraction, which commonly occurs with atrial fibrillation. Cannon a-waves indicate fusion between a- and c-waves culminating in atrial contraction against a closed tricuspid valve, which commonly occurs with atrioventricular dissociation. Cv-wave and the obliteration of x-descent is an indicative of backward flow from the right ventricle through a regurgitant tricuspid valve.¹²

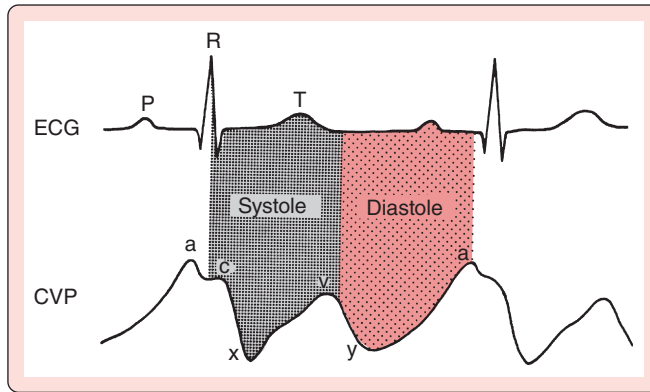


Fig. 4.2 Representative waveform obtained from a central venous pressure (CVP) tracing. (From Mark JB. Central venous pressure monitoring: clinical insights beyond the numbers. *J Cardiothorac Vasc Anesth.* 1991;5:163–173.)

NONINVASIVE AND MINIMALLY INVASIVE METHODS OF HEMODYNAMIC MONITORING

Insertion of invasive arterial monitoring devices or a PAC is not without risk or complication. Accordingly, there is now a greater interest in developing noninvasive and minimally invasive hemodynamic monitoring devices for use in intensive care. There are a number of monitoring modalities currently on the market employed in modern-day ICUs. The use of these devices in any one ICU setting is often predicated on availability and institutional bias. It is beyond the scope of this chapter to exhaustively discuss all available devices. Instead we will address those likely encountered in the intensive care environment.

Pulse Oximetry and Monitoring the Cardiac Cycle

Pulse oximetry is a standard of care monitoring device in the ICU. It is composed of a light-emitting diode sensor that emits red and infrared light to measure oxyhemoglobin and deoxyhemoglobin based on their unique light absorption characteristics. Pulse oximetry monitoring detects changes in hemoglobin oxygen saturation through the use of spectrometry and plethysmography. Pulse oximetry monitoring uses pulsatile flow with each cardiac cycle for the detection of hypoxemia and, therefore, also can serve as a noninvasive assessment of the cardiovascular system.¹³ The pulse oximetry plethysmography waveform is composed of two parts: the large systolic forward flow and small diastolic backward flow. The latter houses a familiar dicrotic notch that echoes the waveform of the arterial blood pressure. Plethysmographic respiratory variation correlates well with pulse pressure variation and is used as a minimally invasive tool to assess volume responsiveness.¹⁴

Capnography and Monitoring of End-Tidal Carbon Dioxide

Capnography measures the amount of carbon dioxide produced in aerobically active tissue delivered by the cardiovascular system to the lungs, where it is then expunged as waste (Fig. 4.3). End-tidal carbon dioxide (ETCO₂), measured by capnography, is the peak partial pressure of carbon dioxide detected in exhalant gases at the point of maximal exhalation. The production of carbon dioxide is a by-product of aerobic respiration, and its presence in the blood is dependent on the ability of the cardiovascular system to transport it to the lungs where it is removed from the body. Assuming that metabolism and ventilation are constant (easily presumed in an ICU with mechanically ventilated patients), ETCO₂ levels indicate the adequacy with which carbon dioxide is carried to the lungs. It is therefore a noninvasive, easily accessible marker of cardiovascular health and cardiac output.

The capnography waveform is composed of four phases. Phase I is flat with a zero value and represents inspiration and early exhaled air from the anatomic dead space. Phase II manifests as a sharply rising slope that represents the transition of exhalant gases arising from anatomic dead space to ventilation occurring at the alveolar level. Phase III, or the alveolar plateau, occurs when exhaled breath arises solely from the alveoli. Phase 0 consists of a rapid decrease in carbon dioxide as freshly inspired gas is drawn across the capnography sensor.

The use of ETCO₂ is often encountered in the hospital setting because it has a myriad of clinical uses. In the cardiac catheterization laboratory it is often employed to monitor patients undergoing moderate sedation. In the surgical suites it is used by anesthesiology as a marker of effective ventilation and, just after intubation, is used to confirm endotracheal tube placement.

The different waveform morphologies produced by continuous ETCO₂ monitoring in the ICU are diagnostic of pathological states. For the purposes of this text, however, we will highlight that capnography and ETCO₂ monitoring guide resuscitation during cardiac arrest by reflecting the effectiveness of cardiopulmonary resuscitation, confirmation of endotracheal tube placement, the return of spontaneous circulation, and termination of resuscitation efforts.¹⁵

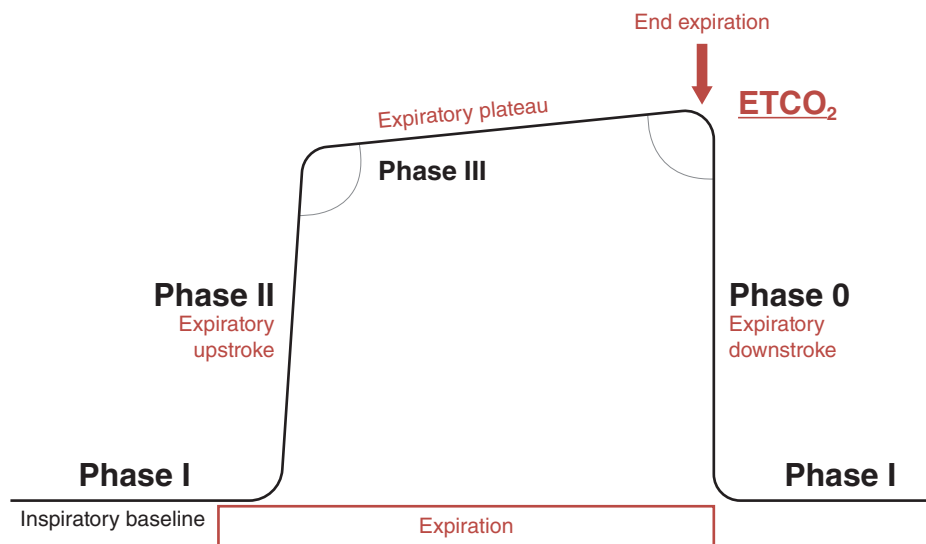


Fig. 4.3 Representative waveform obtained from capnography monitoring. *ETCO₂*, End-tidal carbon dioxide. (From Kerslake I, Kelly F. Uses of capnography in the critical care unit. *BJA Educ.* 2017;17:178–183.)

In mechanically ventilated patients, capnography is paired with the indirect Fick equation to use “partial gas rebreathing” to noninvasively measure cardiac output. At a steady state, the amount of carbon dioxide entering the lungs via the pulmonary artery is proportional to cardiac output and equals the amount exiting the lungs via expiration and pulmonary veins. The concept behind partial gas rebreathing is that when 30 seconds of rebreathing is introduced, the amount of carbon dioxide eliminated by expiration decreases, and the eliminated carbon dioxide increases in proportion to the cardiac output. This method has various limitations including mechanical ventilation and requires the absence of active pulmonary pathology. For these reasons, partial gas rebreathing is limited in its clinical applicability.¹⁶

CONTEMPORARY NONINVASIVE AND MINIMALLY INVASIVE HEMODYNAMIC MONITORING MODALITIES

As previously discussed, canalization of an accessible artery allows for accurate invasive monitoring of arterial blood pressure and MAP. It also has the ability to offer a continuous, minimally invasive method for cardiac output monitoring. Arterial pressure is dependent on stroke volume, central arterial compliance, resistance, and impedance. Several proprietary algorithm models employ these variables to estimate cardiac output from arterial pressure waveform analysis. Devices using these algorithms require an arterial line for their use and, as a result, are referred to as “minimally noninvasive” techniques for assessing cardiac output.

Algorithms developed for estimating cardiac output from arterial access are either calibrated or uncalibrated. Device calibration is important to obtain an initial data set of the arterial system for stroke volume calculation. Calibrated devices require both central venous and arterial catheters and use transpulmonary dilution or esophageal Doppler techniques to calculate a patient's specific compliance and impedance. Calibrated devices require recalibration with major hemodynamic changes. Uncalibrated devices solely require an arterial catheter and use a standard value for compliance and impedance based on the patient's demographic characteristics. Both calibrated and uncalibrated devices that estimate cardiac output from an arterial line are limited when the quality of arterial line waveform analysis is compromised or highly variable. Devices produce inaccurate data when confronted with arrhythmias (atrial fibrillation and atrial flutter), waveform inaccuracies produced by cardiac output augmentation devices (intra-aortic balloon pump), or conditions associated with low vascular tone such as septic shock.

Interrogation of esophageal blood flow using Doppler analysis offers another minimally invasive method for assessing tissue perfusion pressure. With this method, a Doppler probe is placed in the esophagus of a heavily sedated and mechanically ventilated patient. The Doppler probe obtains the flow velocity of blood across the descending aorta and uses proprietary algorithm software to estimate aortic diameter and calculate an estimate of cardiac output. Doppler probes are then either removed or remain in place for a defined period of time (usually less than 48 hours) to guide therapies. Esophageal Doppler monitoring possesses less infectious risk than the presence of indwelling central lines or PACs. In exchange, however, accurate data from esophageal Doppler monitoring is highly dependent on the correct placement of the esophageal probe.

Measurements of electrical bioimpedance across the chest wall offer a safe and noninvasive alternative for estimating cardiac output. Rather than using indwelling catheters to measure cardiac output, electrodes are placed on the chest wall to measure the impedance created by the flow of blood through the aorta (and transmitted across the chest wall). Bioimpedance therefore essentially measures the difference in applied voltage and detected voltage across the chest wall. Volume changes in the aorta will cause changes in applied resistance of a voltage across the chest wall. Changes in impedance over time are used to estimate stroke volume that occurs with the administration of intravenous fluid. This method is highly influenced by the amount of fluid in the thoracic compartment and will produce erroneous data in patients with decompensated heart failure, pleural effusions, or pathology at the aorta.¹⁷ Several devices for measuring bioimpedance are currently available on the market.

DYNAMIC ASSESSMENTS FOR HEMODYNAMIC MONITORING

Up until this point, we have discussed both invasive and noninvasive modalities of hemodynamic monitoring in the intensive care setting. Yet, the reader should note that the use of such monitoring techniques engrains a *reactive*—rather than a *proactive*—approach to patient care. The goal of modern critical care units is to redefine bedside monitoring to identify patients and clinical conditions that might improve with interventions.

Dynamic assessments for the hemodynamic monitoring of critically ill patients generally all attempt to answer the same question: Is it possible to determine where an individual patient's hemodynamics lie on the Frank-Starling curve? Intrinsic to this inquiry is the clinician's desire to not only interrogate cardiovascular physical fitness but to also predict which patients have a reduced cardiac output and will benefit from the delivery of intravenous fluids. These questions are important because the administration of intravenous fluids is not a risk-free endeavor. The prescription of inappropriate intravenous fluids is associated with longer mechanical ventilation times, increased risk of developing renal failure, and higher ICU mortality rates. Determining “fluid responsiveness,” or which patients will improve their cardiac output in response to increased stroke volume, is therefore a more dynamic and forward-thinking approach to bedside care than decisions based on the static and reflexive nature of traditional hemodynamic monitoring.

Pulse pressure and stroke volume variation (PPV and SVV, respectively) play an integral role in the dynamic assessments of fluid responsiveness in critical care. Conceptually, changes in PPV and SVV are based on heart-lung interactions produced by the introduction of positive pressure in the mechanically intubated patient. The effect of adding and removing positive pressure from the respiratory cycle equates to adding and removing preload from the heart. High intrathoracic pressure during positive pressure inspiration limits right ventricular preload and raises afterload by decreasing venous return and increasing transpulmonary pressure. This will lead to a reduction in left ventricular preload and stroke volume causing changes in pulse pressure and stroke volume variation across the respiratory cycle. This phenomenon is called “reverse pulsus paradoxus” because it stands as the mirror opposite to the pulsus paradoxus that occurs in spontaneously breathing patients not receiving mechanical ventilation. An observed exaggeration of reverse pulsus paradoxus (generally reported as a 12%–15% change in PPV or SVV) is seen in patients who are preload responsive and may derive benefit from the administration of intravenous fluids.

Measurement of either PPV or SVV at the bedside is easily accomplished in the ICU by a variety of techniques. Pulse pressure variation is measured by obtaining the range difference between the maximal and minimal values of the pulse pressure and dividing it over the mean of these two values. Current hemodynamic monitors offer the feature of automatic calculation of pulse pressure and stroke volume variation value in patients that have an existing arterial catheter in place. As discussed above, pulse oximetry offers a noninvasive method of visualizing cardiac output. Changes in the amplitude and area under the pulse oximetry waveform correlate with PPV and are also used to identify fluid-responsive patients.

Despite the advantages of using PPV and SVV to recognize fluid-responsive patients, it is also important to recognize the pitfalls and limitations of this technique. PPV and SVV are less reliable and result in false positive values in conditions associated with increased intraabdominal pressure, spontaneous breathing, cardiac arrhythmias, and right ventricular dysfunction. False negative values are common in conditions associated with low tidal volume, low lung compliance, or high respiratory rate (as in patients with acute respiratory distress syndrome).¹⁸

Assessment of respiratory variation at the inferior vena cava (IVC) with mechanical ventilation is another method to assess volume responsiveness (Fig. 4.4). A standard approach to this technique involves the evaluation of the IVC in the subcostal space at the xiphoid process using point-of-care cardiac ultrasound. In this position, the ultrasound probe is used to identify the transition of the IVC into the right atrium. The diameter of the IVC is then measured approximately 2 cm distal to the hepatic vein. During inspiration the diameter of the IVC is assessed for collapsibility. A reduction of IVC diameter by 50% during inspiration corresponds with low right atrial pressures and may indicate a patient who could benefit from augmentation of their stroke volume with the administration of intravenous fluids. This noninvasive technique for monitoring fluid responsiveness is considered a core of most point-of-care ultrasound curricula because it is conducted quickly and safely at the bedside with high reproducibility across trained clinicians.

The dynamic monitoring maneuvers outlined above offer a “before and after” snapshot of cardiac output to identify patients that are “preload responsive.” While intravenous fluids can be administered to obtain this data, alternatives to the administration of intravenous fluids are also paired with these monitoring techniques with good results. For example, passively raising a patient's legs from the supine position to a 45-degree angle has effects similar to the administration of a 250- to 500-mL bolus of intravenous fluids. A 10% change in cardiac output (usually assessed by changes in MAP, stroke volume, or pulse pressure) equates to a preload-responsive patient. Though perhaps of limited

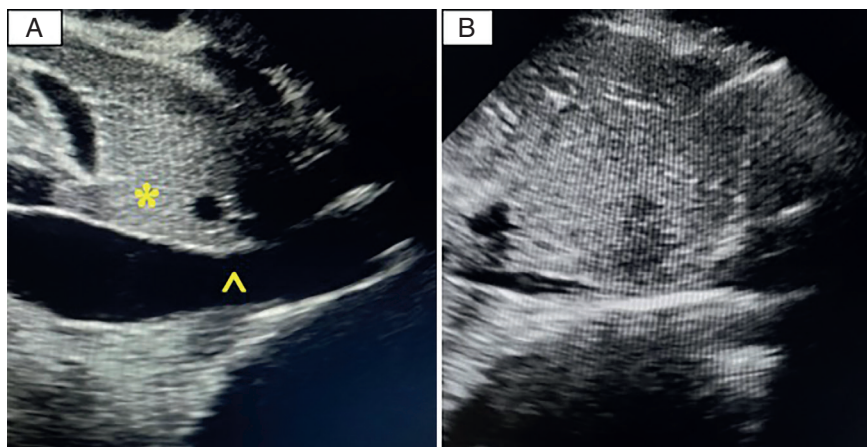


Fig. 4.4 Ultrasonographic measurement of inferior vena cava (IVC) compressibility for measurement of volume responsiveness. (A) Ultrasound imaging of the inferior vena cava (^) is demonstrated as it transitions from the liver (*) to the right atrium. (B) Effects of inspiration on a collapsible IVC. The change in IVC diameter by 50% is associated with the theoretical increase in cardiac output with the hypothetical administration of intravenous fluids.

utility in obese patients or patients with intraabdominal hypertension, the passive straight leg test has the advantage of equating to an “internal” bolus without influencing true volume status. Similarly it is estimated that temporarily occluding an artificial airway for 15 seconds at the end of expiration can also offer insight into volume responsiveness. This method takes advantage of the increase in intrathoracic pressures that occur during inspiration in patients receiving positive-pressure mechanical ventilation. An increase in cardiac output with this maneuver by 5% implies that a patient is fluid responsive.¹⁹

While there is a growing body of evidence to support the use of fluid responsiveness as a dynamic hemodynamic monitoring technique in unstable patients, the lack of robust adaptation of these philosophies likely involves a myriad of reasons. Many techniques to evaluate dynamic hemodynamic monitoring require advanced skill sets or imaging modalities (such as ultrasound). Furthermore, there is a lack of randomized control trials comparing various measurements of fluid-responsive techniques to justify the use of any one modality. Nevertheless, the use of fluid responsiveness to monitor bedside hemodynamics represents a potential paradigm shift away from classical, static hemodynamic monitoring to more novel dynamic techniques for the betterment of patient care.

CONCLUSION

Monitoring the critically ill patient in the setting of the ICU is a complex and detail-oriented endeavor. It is not unusual to observe patients in these environments suffering from multiorgan failure that evolves in varying degrees over time. It is of the utmost importance for clinicians to possess competency in various bedside monitoring techniques for the early recognition of cardiac insufficiency. Fortunately, much of the data collected from bedside intensive monitoring is rooted in familiar concepts of cardiovascular and pulmonary physiology. Therefore, bedside monitoring possesses high-yield clinical information and relevance. Exploring the use of specific monitoring devices, their operational benefits, their limitations, and their contribution to the development of intuitional treatment approaches are essential for the delivery of intensive care medicine by every bedside practitioner.

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